

Rahul Rawat

MASTER THESIS STUDENT

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PROFILE

Data Science and Analytics Master student at SRH Heidelberg based in Mannheim, with practical experience building and evaluating Generative AI systems on real data. I shipped a modular RAG orchestrator on Llama 3.1 with LangChain that classifies user intent across retrieval, web search, and conversation, and I designed a fairness evaluation and bias detection framework for CreditIQ that lifted the Disparate Impact ratio from 0.79 to 0.88. My Bachelor Thesis published as an IEEE style paper compares six classifiers on a clinical dataset with 10 fold cross validation and ROC AUC as the headline metric, which is the same evaluation discipline that an automated Gen AI evaluation thesis needs.

EXPERIENCE

eRay GmbH

Oct 2025 to Mar 2026

Data Scientist

- Built an end to end recursive time series pipeline forecasting chlorophyll a, turbidity, pH, and dissolved oxygen for a German lake, delivered as a six month collaboration with SRH University Heidelberg.
- Benchmarked six models head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and used CatBoost multi quantile regression to produce asymmetric 80 percent prediction intervals.
- Enforced strict anti leakage rules across the pipeline, surfacing the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors.
- Reconstructed missing winter readings with MICE imputation and engineered a synthetic winter decay forecast canvas so tree based models stopped flatlining during recursive prediction.
- Wrapped the whole pipeline in an orchestrator with gate checks and velocity and ecological bounds that halts on failed imputation rather than letting bad data cascade downstream.

EDUCATION

M.Sc. Data Science and Analytics

Apr 2025 to Present

SRH University of Applied Sciences Heidelberg

Bachelor of Technology in Computer Science

2019 to 2023

GL Bajaj Institute of Technology and Management

PERSONAL PROJECTS

Hybrid RAG Orchestrator

Python

LangChain

Llama 3.1 8b on Groq

ChromaDB

HuggingFace MiniLM L6 v2

Streamlit

- Designed a modular Retrieval Augmented Generation system using Llama 3.1 8b via Groq for high speed inference and LangChain for orchestration across multiple execution paths.
- Engineered a custom decision making router that dynamically classifies user intent into three execution paths, local knowledge retrieval on PDF and vector stores, external web search through DuckDuckGo, and direct conversational logic.
- Implemented ChromaDB with local persistence for document storage, using HuggingFace MiniLM L6 v2 for semantic embeddings tuned for retrieval quality.
- Built a stateful MemoryAgent that maintains conversational context across multiple turns and integrated it directly into the inference pipeline for coherent multi turn behaviour.
- Shipped the whole system behind a Streamlit interface as a deployed end to end tool, owning the design, implementation, and rollout in full.

CreditIQ, Fairness by Design Credit Scoring System

Python

scikit learn

AIF360

SHAP

Streamlit

- Lifted the model's Disparate Impact ratio from a failing 0.79 to a compliant 0.88, clearing the EU AI Act and AGG 80 percent fairness threshold.
- Diagnosed a hidden intersectional bias where younger women were still being penalised even after single axis age bias was fixed, using SHAP, then designed a four way age by gender threshold matrix to correct it without over correcting into reverse discrimination.
- Cut the false negative rate from 44 percent to 16.7 percent while holding accuracy at 75 percent, documenting the fairness accuracy trade off as a deliberate and regulator defensible decision.
- Shipped a Streamlit decision support tool giving finance managers a recommendation plus a plain language LLM generated explanation, keeping a human in the loop per GDPR Article 22 and EU AI Act Article 14.
- Backed the pipeline with unit tests at 100 percent branch coverage and a full regulatory write up covering EU AI Act Annex III, GDPR, a model card, and attack vectors.

Movie Analytics and ML Pipeline, Cloud Native Data Platform

GCP BigQuery

Cloud Run

GCS

Cloud Scheduler

BigQuery ML

Python

SQL

Looker Studio

- Built an end to end batch pipeline pulling movie data from a public API into a GCS data lake, processed through Bronze to Silver to Gold medallion architecture in BigQuery, fully automated on a Cloud Scheduler trigger with no manual steps.
- Engineered the Silver layer for trustworthy analytics with schema enforcement, safe type casting, deduplication via window functions, and genre normalisation into a relational model.
- Trained a BigQuery ML classifier to predict whether a film will be a hit before release, deliberately splitting features into two tables to prevent data leakage and keeping only pre release signals.
- Built Gold layer aggregates and a five page Looker Studio dashboard answering concrete business questions on genre ROI, foreign language growth, and release season timing, plus an ML early warning view.
- Secured the system with a least privilege service account and Secret Manager and version controlled all code, SQL, and schemas in a reproducible GitHub repository.

RESEARCH AND THESIS

Bachelor Thesis, Diabetes Prediction Using Machine Learning

GL Bajaj Institute of Technology and Management, IEEE style paper

- Built a full end to end machine learning pipeline comparing six classifiers on a clinical diabetes dataset of 768 patients, using 10 fold cross validation and per model confusion matrices.
- Caught biologically impossible zero values that the original authors had overlooked, then applied IQR based outlier removal and proper imputation to restore data integrity.
- Chose ROC AUC over accuracy as the headline metric for a 65 to 35 imbalanced dataset, avoiding a misleadingly rosy accuracy figure that would have hidden real error patterns.
- Wrote up findings as an IEEE style paper including an honest section on limitations and what to do differently in a follow up study.

LANGUAGES

English: fluent, professional working proficiency. **German:** B1, in progress.